i. Extrinsic factors

- Drugs and chemicals
- Infection
- Antibody reaction

ii. Intrinsic factors

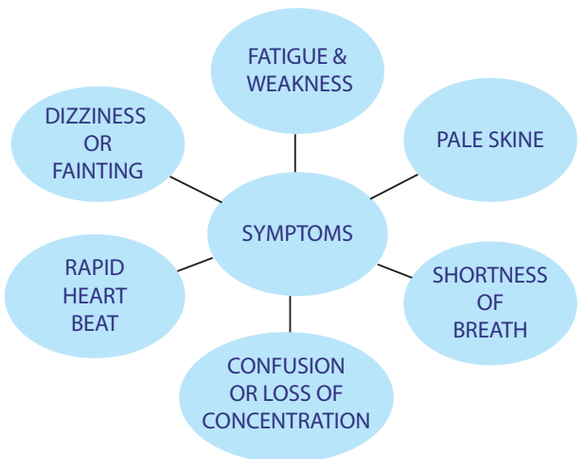
- Abnormalities of RBC membrane
- Abnormal haemoglobin synthesis sickle cell, disease, thalasemia syndrome

Pathophysiology

RBCs and haemoglobin are normally formed at the same rate at which they are destroyed.

Whenever formation of RBCs or haemoglobin is decreased or their destruction is increased, anaemia results.

Signs and Symptoms



Diagnosis

1. Complete blood count
2. Haemoglobin electrophoresis

Types of Anaemia

1. **Anaemia from blood loss:** Haemorrhage or continued slow bleeding will cause anaemia.
2. **Iron deficiency Anaemia:** It may result from faulty eating habits (Poor diets or hurrying meals).
3. **Pernicious Anaemia:** A patient with pernicious Anaemia lacks substances in the gastric juice called instrinsic factor, which is necessary to enable the body to absorb vitamin B_{12} from food.
4. **Sickle cell anaemia:** It is genetic disease in which the red blood cells become sickle in shape due to the presence of abnormal haemoglobin.
5. **Aplastic anaemia:** It results from disease of the bone marrow (where most blood cells are produced.) whereby the marrow is destroyed.
6. **Megaloblastic anaemia:** It is caused by deficiency of the vitamin B_{12} and folic acid. This shows identical bone marrow and peripheral blood changes, because both vitamins are essential for normal DNA synthesis.
7. **Haemolytic anaemia:** The erythrocyte has a shortened life span.

Management

1. **Anaemia from blood loss:** Replace the blood cells by transfusion and at times administering iron supplements.
2. **Iron-Deficiency Anaemia:** Take extra iron containing food.
3. **Pernicious Anaemia:** Take vitamin B_{12} containing foods and injection every 2 or 3 weeks will allow the person to live normally.
4. **Sickle cell Anaemia:** Citadels citrate, pentoxifylline, vanillin oil as anti sickly

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